Constraint-Aware Hierarchical Risk Parity for Robust Portfolio Allocation

Pramukh Rajendra Jain
School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India
pramukhjain.9966@gmail.com

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Abstract

Hierarchical Risk Parity (HRP) is a popular portfolio construction method that allocates capital through a correlation tree instead of inverting a covariance matrix, which makes it robust to estimation error. However, HRP cannot respect the constraints that real mandates impose: per-asset limits, sector or asset-class caps and floors, and turnover budgets. We propose Constraint-Aware HRP (CA-HRP), which keeps HRP's hierarchical risk allocation as a target and maps it to the feasible portfolio with minimum ex-ante tracking error to it. We prove that this projection is unique, leaves compliant HRP portfolios unchanged, and redistributes excess weight as a covariance-based hedge. We also introduce a solver-free capacity-aware bisection and prove that it always meets per-asset bounds exactly. In a walk-forward study on 20 multi-asset ETFs (2008–2026), 43 NIFTY 50 stocks (2011–2026) and 47 US large caps (2006–2026), with the design chosen on a validation period and a held-out test period, CA-HRP satisfies every constraint at every rebalance, whereas HRP breaches them at up to 100% of rebalances and the common clip-and-renormalise fix at up to 46%. Where constraints bind, CA-HRP raises the Sharpe ratio from 0.76 to 0.91 relative to Euclidean projection, with 25% lower volatility and smaller drawdowns, and is competitive with constrained minimum variance while producing more stable weights. Where constraints rarely bind, it matches HRP. An ablation shows that the risk-aware projection metric, not the choice of estimator or clustering rule, drives the gains.

Keywords: hierarchical risk parity, portfolio optimisation, constrained allocation, tracking error, robust estimation, convex optimisation.

1 Introduction

Mean–variance optimisation [9] is the textbook answer to portfolio construction, yet in practice its outputs are notoriously unstable: small errors in estimated means and covariances are amplified by the inversion of an ill-conditioned covariance matrix into extreme, concentrated and rapidly changing weights [1, 10]. Out of sample, optimised portfolios frequently fail to beat naive equal weighting [2]. Two families of remedies have emerged. The first regularises the inputs, for example by shrinking the covariance matrix [5] or by imposing weight constraints, which Jagannathan and Ma [4] show act as a form of shrinkage. The second abandons optimisation altogether in favour of heuristic risk allocation, such as equal risk contribution [8].

Hierarchical Risk Parity (HRP), introduced by López de Prado [7], belongs to the second family and has become one of the most popular allocation methods among practitioners and in open-source libraries. HRP clusters assets by correlation, orders them so that similar assets are adjacent, and allocates capital top-down by recursively splitting each cluster in inverse proportion to its risk. Because it never inverts the covariance matrix, HRP is robust to estimation noise and can handle singular covariance matrices.

HRP has, however, a practical blind spot: it cannot respect constraints. Real mandates impose per-asset limits (UCITS concentration rules, single-stock caps), sector or asset-class caps and

floors (investment policy statements), and turnover budgets (to control trading costs). Recursive bisection has no mechanism for any of these, and in our multi-asset universe unconstrained HRP places on average 86% of capital in a single short-term bond fund, breaching a 15% single-fund cap and a 40% asset-class cap at *every* rebalance. The usual fixes are ad hoc: weights are clipped and renormalised, which can cycle without ever reaching the feasible set, or projected onto it with a Euclidean distance that ignores how assets co-move.

This paper proposes **Constraint-Aware Hierarchical Risk Parity (CA-HRP)**, a principled way to keep HRP's risk allocation while satisfying any set of linear constraints exactly. Our contributions are:

1. **Capacity-aware recursive bisection.** We define a capacity interval for every node of the HRP tree and clip each split into it. We prove that the interval is never empty and that per-asset bounds are met exactly, with no solver (Proposition 1). This yields a solver-free variant, CA-HRP-B.

2. **Tracking-error projection.** We map the HRP target to the feasible portfolio with minimum ex-ante tracking error to it. We show the projection is unique, leaves compliant HRP portfolios unchanged, and redistributes excess weight as a covariance-based hedge rather than uniformly (Proposition 2).

3. **A transparent evaluation protocol.** The method design was selected on a validation period only and then frozen; we report a separate held-out test period, all constraint sets were fixed before any backtest was run, and all code and data are public.

4. **Evidence on three universes.** On 20 multi-asset ETFs (2008–2026), 43 NIFTY 50 stocks (2011–2026) and 47 US large caps (2006–2026), CA-HRP satisfies every constraint at every rebalance. Where constraints bind, it is the strongest member of the HRP family and is competitive with constrained minimum variance; where they rarely bind, it matches HRP.

The rest of the paper is organised as follows. Section 2 reviews related work, Section 3 states the problem, Section 4 presents the method and its properties, Section 5 describes the data and experimental protocol, Section 6 reports results, and Section 8 concludes.

2 Related Work

Estimation error and robust allocation. The sensitivity of mean–variance portfolios to input error is well documented [1, 10]. DeMiguel et al. [2] find that across many data sets none of fourteen optimised models consistently beats $1/N$ out of sample. Shrinkage estimators [5] reduce the condition number of the covariance matrix, and Jagannathan and Ma [4] show that no-short-sale constraints behave like covariance shrinkage. Risk-based methods allocate by risk rather than expected return; the equal-risk-contribution (ERC) portfolio [8] equalises marginal risk contributions and can be computed by convex optimisation [16].

Hierarchical allocation. López de Prado [7] introduced HRP, combining graph-theoretic clustering with recursive inverse-variance bisection. Raffinot [13] studied hierarchical clustering-based allocation more broadly, and Raffinot [14] proposed the Hierarchical Equal Risk Contribution (HERC) portfolio, which splits along the dendrogram rather than the ordered list. Pfitzinger and Katzke [11] recognised that HRP lacks constraint handling and proposed a cluster-based capital allocation that accommodates weight bounds within the bisection.

Positioning of this work. Capacity-aware bisection is closest in spirit to Pfitzinger and Katzke [11]; our contribution is a capacity definition that also incorporates group caps, together with a short proof of non-emptiness and exact box feasibility based on the additivity and sub-additivity of the capacity bounds. More importantly, we show empirically that enforcing constraints *inside*

the bisection has a hidden cost: excess weight cascades down one branch and can concentrate in volatile assets. Our main method therefore separates the two concerns: HRP supplies the risk allocation, and a tracking-error projection [15] enforces arbitrary linear constraints (group floors, caps and turnover) with minimum risk distortion. To our knowledge this is the first study to compare constraint-handling strategies for HRP under a common, pre-registered protocol with a held-out test period.

3 Problem Setting

We consider N risky assets with daily simple returns $r_t \in \mathbb{R}^N$. At each rebalancing date the manager observes the last L return vectors, estimates a covariance matrix $\hat{\Sigma}$, and chooses weights $w \in \mathbb{R}^N$. The mandate imposes a polyhedral feasible set

$$\mathcal{W}(w^-) = \Big\{ w : \ \mathbf{1}^\top w = 1, \ \ \ell \le w \le u, \ \ \underline{c}_g \le \sum_{i \in g} w_i \le \bar{c}_g \ \ \forall g \in \mathcal{G}, \ \ \|w - w^-\|_1 \le \tau \Big\}, \tag{1}$$

where ℓ, u are per-asset bounds, $\mathcal{G}$ partitions the assets into groups (sectors or asset classes) with caps $\bar{c}_g$ and floors $\underline{c}_g$, w^- is the drifted pre-trade portfolio and τ a turnover budget. All constraints are linear, so $\mathcal{W}$ is a convex polytope. We assume it is non-empty. The turnover budget can in principle conflict with the other constraints if price drift pushes the pre-trade portfolio far outside the box or group limits; our implementation then relaxes the turnover cap for that date only. In all experiments reported here this fallback was never triggered: every CA-HRP portfolio satisfied all constraints, turnover included.

4 Method

4.1 Hierarchical Risk Parity

HRP [7] proceeds in three steps. (i) *Tree clustering*: the correlation matrix ρ implied by $\hat{\Sigma}$ is mapped to the distance $d_{ij} = \sqrt{(1-\rho_{ij})/2}$ and agglomerated with single linkage. (ii) *Quasi-diagonalisation*: assets are ordered by the dendrogram leaves so that similar assets are adjacent. (iii) *Recursive bisection*: starting from the full ordered list with weight $W = 1$, every cluster C is split into a left and right half C_L, C_R; with $V_S = \tilde{w}_S^\top \hat{\Sigma}_{SS} \tilde{w}_S$ the variance of the inverse-variance portfolio $\tilde{w}_S \propto \mathrm{diag}(\hat{\Sigma}_{SS})^{-1}$ of a cluster S, the split fraction is

$$\alpha_C = 1 - \frac{V_{C_L}}{V_{C_L} + V_{C_R}}, \qquad W_{C_L} = \alpha_C W_C, \quad W_{C_R} = (1-\alpha_C) W_C. \tag{2}$$

The recursion ends at single assets. No matrix is inverted and every split is a two-asset risk-parity decision, which is the source of HRP's stability. Nothing in (2), however, knows about $\mathcal{W}$.

4.2 Capacity-aware recursive bisection (CA-HRP-B)

Our first contribution makes the bisection itself respect per-asset bounds and, partially, group caps, without any optimisation solver. For a set of assets S define its *capacity interval*

$$\mathrm{lo}(S) = \sum_{i \in S} \ell_i, \qquad \mathrm{hi}(S) = \sum_{g \in \mathcal{G}} \min\Big(\sum_{i \in S \cap g} u_i, \ \bar{c}_g \Big). \tag{3}$$

$\mathrm{lo}(S)$ is the least weight that S must hold and $\mathrm{hi}(S)$ an upper bound on what it can hold. When a cluster C carrying weight W_C is split, the HRP fraction (2) is clipped to

$$\alpha_C \in \Big[\tfrac{\max(\mathrm{lo}(C_L),\, W_C - \mathrm{hi}(C_R))}{W_C}, \ \ \tfrac{\min(\mathrm{hi}(C_L),\, W_C - \mathrm{lo}(C_R))}{W_C} \Big]. \tag{4}$$

Proposition 1 (Exact box feasibility)**.** *If* $\mathrm{lo}(\{1,\dots,N\}) \le 1 \le \mathrm{hi}(\{1,\dots,N\})$*, the interval* (4) *is non-empty at every split, and the resulting weights satisfy* $\mathbf{1}^\top w = 1$ *and* $\ell_i \le w_i \le \min(u_i, \bar{c}_{g(i)})$ *for all* i*.*

Proof. We show by induction that every cluster satisfies the invariant $\mathrm{lo}(C) \le W_C \le \mathrm{hi}(C)$. It holds at the root by assumption. Note that lo is additive, $\mathrm{lo}(C) = \mathrm{lo}(C_L) + \mathrm{lo}(C_R)$, and hi is sub-additive because $\min(a+b, c) \le \min(a, c) + \min(b, c)$ for $a, b, c \ge 0$, so $\mathrm{hi}(C) \le \mathrm{hi}(C_L) + \mathrm{hi}(C_R)$. The interval (4) (multiplied by W_C) is non-empty iff the four inequalities $\mathrm{lo}(C_L) \le \mathrm{hi}(C_L)$, $\mathrm{lo}(C_L) \le W_C - \mathrm{lo}(C_R)$, $W_C - \mathrm{hi}(C_R) \le \mathrm{hi}(C_L)$ and $W_C - \mathrm{hi}(C_R) \le W_C - \mathrm{lo}(C_R)$ hold. The first and last follow from $\ell \le u$; the second from $W_C \ge \mathrm{lo}(C)$ and additivity; the third from $W_C \le \mathrm{hi}(C)$ and sub-additivity. Any α_C in the interval gives children with $\mathrm{lo}(C_L) \le W_{C_L} \le \mathrm{hi}(C_L)$ and likewise for C_R, which preserves the invariant. At a leaf $\{i\}$, $\mathrm{lo} = \ell_i$ and $\mathrm{hi} = \min(u_i, \bar{c}_{g(i)})$, which proves the claim. Weights sum to one because each split conserves W_C. □

CA-HRP-B costs $O(N^2)$ beyond HRP and needs no solver. It guarantees box bounds exactly, but group constraints are only respected when a group lies inside one sub-tree, and group floors and turnover are not handled. Those remaining constraints are enforced by a light projection (next subsection, with the Euclidean metric replaced by a tree-weighted one; details in the code). As our experiments show, the price of handling constraints *inside* the bisection is that excess weight cascades down a single branch of the tree and can end up concentrated in volatile assets; this motivates the main method.

4.3 Tracking-error projection (CA-HRP)

The key observation is that HRP produces an excellent *target* but has no notion of distance between portfolios. Practitioners close the gap by clipping and renormalising, which is not guaranteed to reach $\mathcal{W}$, or by a Euclidean projection $\min_{w \in \mathcal{W}} \|w - w^{\mathrm{HRP}}\|_2^2$, which treats a unit of weight moved from a short-term bond fund to an oil fund the same as one moved to another bond fund. We instead measure distance in *risk*: the ex-ante tracking-error variance of the constrained portfolio relative to HRP [15],

$$w^{\mathrm{CA}} = \arg\min_{w \in \mathcal{W}(w^-)} (w - w^{\mathrm{HRP}})^\top \bar{\Sigma}\, (w - w^{\mathrm{HRP}}), \qquad \bar{\Sigma} = \frac{N\hat{\Sigma}_{\mathrm{LW}}}{\operatorname{tr} \hat{\Sigma}_{\mathrm{LW}}} + \epsilon I, \tag{5}$$

where $\hat{\Sigma}_{\mathrm{LW}}$ is the Ledoit–Wolf shrinkage estimator [5], w^{HRP} is HRP computed on $\hat{\Sigma}_{\mathrm{LW}}$, and the small ridge $\epsilon = 10^{-4}$ guarantees strict convexity.

Proposition 2 (Properties of the projection)**.** *(a) If* $\mathcal{W}(w^-) \neq \emptyset$*, problem* (5) *has a unique solution. (b) If* $w^{\mathrm{HRP}} \in \mathcal{W}(w^-)$ *then* $w^{\mathrm{CA}} = w^{\mathrm{HRP}}$*: CA-HRP never alters HRP when HRP already complies. (c) Among all feasible portfolios,* w^{CA} *minimises the ex-ante variance of the return difference to HRP, so it is the feasible portfolio whose returns are most correlated with, and least volatile around, the HRP portfolio. (d) Let* $Aw \le b$ *collect the active inequality constraints and* $\mathbf{1}^\top w = 1$ *the budget. The KKT conditions give* $w^{\mathrm{CA}} = w^{\mathrm{HRP}} - \frac{1}{2}\bar{\Sigma}^{-1}(A^\top \lambda + \nu \mathbf{1})$ *with* $\lambda \ge 0$*: the correction is a* hedge*. Weight removed from a capped asset is redistributed in proportion to the rows of* $\bar{\Sigma}^{-1}$*, i.e. towards the assets that best replicate the capped asset's risk.*

Proof. (a) $\bar{\Sigma} \succ 0$, so the objective is strictly convex and $\mathcal{W}$ is a non-empty closed convex set; a unique minimiser exists. (b) The objective is non-negative and zero at w^{HRP}. (c) $(w - w^{\mathrm{HRP}})^\top \hat{\Sigma}(w - w^{\mathrm{HRP}})$ is the variance of $(w - w^{\mathrm{HRP}})^\top r$ under the estimated covariance; the trace normalisation and ridge only rescale and regularise it. (d) Stationarity of the Lagrangian $(w - w^{\mathrm{HRP}})^\top \bar{\Sigma}(w - w^{\mathrm{HRP}}) + \lambda^\top (Aw - b) + \nu(\mathbf{1}^\top w - 1)$, with the ℓ_1 turnover constraint written in its standard linear epigraph form. □

Part (d) explains the behaviour we observe in the multi-asset universe: when HRP wants far more than the 40% asset-class cap in bonds, the Euclidean projection spreads the excess almost

Algorithm 1 Constraint-Aware HRP (one rebalancing date)

Require: returns window $R \in \mathbb{R}^{L \times N}$, feasible set $\mathcal{W}$, drifted weights w^-

1: $\hat{\Sigma}_{\mathrm{LW}} \leftarrow$ Ledoit–Wolf shrinkage covariance of R
2: $\rho \leftarrow$ correlation of $\hat{\Sigma}_{\mathrm{LW}}$; $d_{ij} \leftarrow \sqrt{(1-\rho_{ij})/2}$
3: $\pi \leftarrow$ leaf order of the single-linkage dendrogram of d
4: $w^{\mathrm{HRP}} \leftarrow$ recursive bisection (2) on π ▷ optionally clipped by (4)
5: $w^{\mathrm{CA}} \leftarrow$ solve the convex QP (5) over $\mathcal{W}(w^-)$
6: **return** w^{CA}

uniformly (about 5% to each non-bond fund), while the tracking-error projection re-shapes the portfolio as a hedge: inside the bond budget it moves from short-term to intermediate and long Treasuries (IEF, TLT), whose negative correlation with equities offsets the risk of the assets that must now be held, and outside the bond budget it favours gold and broad commodities (Section 6). The constrained portfolio keeps the defensive character of HRP, with markedly lower volatility than the Euclidean alternative.

Complexity. Steps 1–4 cost $O(LN^2 + N^2 \log N)$. Step 5 is a QP with N variables (plus N auxiliary variables for the turnover epigraph) and $N + |\mathcal{G}|$ linear constraints; with an interior-point solver [3] it takes a few milliseconds for $N \leq 50$. A full 20-year monthly backtest of CA-HRP runs in under six seconds on a laptop.

5 Experimental Design

Data. We use daily dividend- and split-adjusted closing prices from Yahoo Finance for three universes chosen to span different constraint regimes (Table 1): (i) 20 US-listed multi-asset ETFs covering equities, government and corporate bonds, commodities and real estate (January 2008 – October 2026); (ii) 43 long-listed constituents of India's NIFTY 50 index across 12 sectors (January 2011 – October 2026); (iii) 47 US large-cap stocks across 11 sectors (January 2006 – October 2026). All series are complete over their sample. Survivorship bias affects the stock universes, which consist of current large caps; it affects all methods equally and does not change their ranking, which is our object of study.

Table 1: Universes and constraint sets. Constraint sets were fixed before any backtest.

	Multi-asset ETFs	NIFTY 50 stocks	US large caps
Assets / groups	20 / 4	43 / 12	47 / 11
Sample	2008–2026	2011–2026	2006–2026
Per-asset cap u_i	15%	6%	6%
Group cap $\bar{c}_g$	40%	25%	20%
Group floor $\underline{c}_g$	Equity $\geq$ 20%	–	–
Turnover budget τ (per month)	30%	30%	30%
Validation / test split	2016 | 2017–	2018 | 2019–	2015 | 2016–

Constraints. The constraint sets mirror typical mandates: a single-holding cap of roughly $2.5/N$, sector or asset-class caps, a growth floor of 20% in equities for the multi-asset portfolio, and a monthly two-sided turnover budget of 30%. All weights are long-only and fully invested.

Methods compared. *Unconstrained references*: equal weight (EW), inverse variance (IVP), long-only minimum variance (MinVar), equal risk contribution (ERC) and standard HRP. *Constrained competitors*: minimum variance (MinVar-C) and mean–variance with risk aversion 5

(MVO-C), both solved over the full feasible set; HRP with iterative clipping and renormalisation (HRP-Clip, the common practitioner fix, which ignores turnover); and HRP followed by Euclidean projection onto the full feasible set (HRP-Proj). *Proposed*: the solver-free capacity-aware bisection (CA-HRP-B) and the full method with tracking-error projection (CA-HRP). Covariance-based competitors use the Ledoit–Wolf estimator, except the HRP baselines, which follow the original sample-covariance recipe.

Backtest protocol. At each month end we estimate inputs from the previous $L = 252$ trading days, compute target weights, and trade at the close. Between rebalances weights drift with prices. Turnover is $\|w_t - w_t^-\|_1$ against the drifted weights, and a cost of 10 bps per unit of turnover is deducted (we also report 0, 25 and 50 bps). Constraint compliance is checked on the target weights at every rebalance.

Design selection and held-out test. An initial version of CA-HRP enforced all constraints inside the bisection followed by a tree-weighted projection. Before finalising the method, we compared eight configurations *on the validation period only* (first part of each sample, Table 1) and selected the tracking-error projection with Ledoit–Wolf covariance (Appendix A). The configuration was then frozen, and we report results both over the full sample and over the untouched test period.

Metrics. Annualised return (CAGR), volatility, Sharpe ratio (zero risk-free rate), maximum drawdown, daily 5% CVaR, annualised turnover, constraint-violation rate (share of rebalances with any breach), realised tracking error to unconstrained HRP ($\mathrm{TE}_{\mathrm{HRP}}$), effective number of holdings $1/\sum_i w_i^2$, and risk concentration, the Herfindahl index of percentage risk contributions scaled by N (1 = equal risk). Sharpe ratio differences are tested with a circular block bootstrap (block of 21 days, 2,000 replications) in the spirit of Ledoit and Wolf [6] and Politis and Romano [12].

6 Results

6.1 Constraint compliance

Table 2 first answers the practical question. Unconstrained HRP breaches the mandate at 100% of rebalances in the multi-asset universe, 48% for NIFTY and 66% for US stocks. The common practitioner fix, HRP-Clip, still breaches constraints at 46%, 8% and 8% of rebalances respectively: alternating clipping and group rescaling can cycle when per-asset and group limits interact, and it ignores turnover entirely. Every optimisation-based method, including both CA-HRP variants, satisfies all constraints at every rebalance, and CA-HRP never needed the turnover fallback.

6.2 Performance where constraints bind

The multi-asset universe is where constraints bite hardest: HRP would hold on average 86% in a single short-term Treasury fund (Figure 2), so the 15% fund cap, the 40% asset-class cap and the 20% equity floor force a very different portfolio. Here the way the constraints are enforced matters a great deal. CA-HRP reaches a Sharpe ratio of 0.908 with 7.4% volatility and a −18.8% maximum drawdown, against 0.764, 9.9% and −22.8% for the Euclidean projection (HRP-Proj) and 0.709 for HRP-Clip. The difference with HRP-Clip is significant ($p = 0.01$); against HRP-Proj and equal weight the p-values are 0.13 and 0.12. Among all constrained methods, only constrained minimum variance has a higher Sharpe ratio (0.935), a difference that is far from significant ($p = 0.34$). CA-HRP's realised tracking error to HRP (6.4%) is the lowest of the HRP-family methods, as its construction intends. The composition explains why. Averaged over all rebalances, HRP-Proj spreads the weight it removes from bonds almost evenly (about 5% in each non-bond fund), whereas

Table 2: Full-sample out-of-sample results, net of 10 bps costs. CAGR, Vol, MaxDD, CVaR and TE_{HRP} in %; turnover annualised; Viol = share of rebalances breaching any constraint; p = bootstrap p-value of the Sharpe difference with CA-HRP. Unconstrained methods (top block) are shown for reference only. Best constrained Sharpe ratio in bold.

Universe	Method	CAGR	Vol	Sharpe	p	MaxDD	$CVaR_5$	Turn.	Viol	TE_{HRP}	Eff. N
Multi-asset ETFs	EW	8.0	10.9	0.763	0.12	-24.8	-1.63	0.34	0%	10.2	20.0
	IVP	1.9	2.2	0.840	0.70	-9.6	-0.34	0.30	100%	0.5	1.6
	MinVar	2.4	2.4	0.971	0.68	-9.1	-0.36	0.90	100%	1.4	2.5
	ERC	4.8	5.7	0.852	0.47	-17.3	-0.84	0.48	100%	4.5	9.3
	HRP	1.6	2.0	0.833	0.70	-8.8	-0.30	0.63	100%	0.0	1.4
	MinVar-C	6.8	7.3	**0.935**	0.34	-17.5	-1.07	1.21	0%	6.5	8.0
	MVO-C	8.8	12.2	0.753	0.30	-19.1	-1.91	3.16	0%	11.6	7.3
	HRP-Clip	6.3	9.2	0.709	0.01	-22.8	-1.37	2.17	46%	8.4	13.9
	HRP-Proj	7.4	9.9	0.764	0.13	-22.8	-1.49	0.62	0%	9.3	16.1
	CA-HRP-B	7.1	9.8	0.750	0.14	-21.4	-1.47	2.50	0%	9.2	10.7
	CA-HRP	6.6	7.4	0.908	–	-18.8	-1.08	1.25	0%	6.4	8.3
NIFTY 50 stocks	EW	19.7	15.4	1.247	0.07	-36.0	-2.21	0.62	0%	3.7	43.0
	IVP	18.7	14.3	1.272	0.02	-33.4	-2.04	0.85	17%	2.0	37.8
	MinVar	18.0	12.8	1.356	0.89	-26.1	-1.74	2.87	100%	4.7	14.1
	ERC	19.5	14.2	1.326	0.22	-32.7	-2.02	0.81	3%	1.8	39.0
	HRP	19.5	13.7	1.368	0.96	-30.6	-1.94	2.70	48%	0.0	32.4
	MinVar-C	18.6	12.8	**1.401**	0.62	-26.5	-1.76	2.12	0%	3.5	19.8
	MVO-C	20.2	16.2	1.216	0.17	-33.2	-2.35	3.35	0%	6.8	17.4
	HRP-Clip	19.6	13.7	1.369	1.00	-30.6	-1.95	2.66	8%	0.2	32.8
	HRP-Proj	19.6	13.8	1.368	0.97	-30.6	-1.95	2.62	0%	0.2	32.9
	CA-HRP-B	19.5	13.7	1.368	0.68	-30.1	-1.95	2.51	0%	0.9	33.1
	CA-HRP	19.5	13.7	1.369	–	-30.1	-1.94	2.47	0%	0.9	33.1
US large caps	EW	13.3	18.5	0.767	0.59	-47.9	-2.80	0.53	0%	4.8	47.0
	IVP	11.7	16.1	0.768	0.15	-41.1	-2.40	0.73	24%	1.3	37.0
	MinVar	9.3	13.7	0.720	0.39	-33.0	-1.98	3.02	100%	6.2	10.5
	ERC	12.4	16.8	0.781	0.67	-44.7	-2.54	0.68	3%	2.3	42.6
	HRP	11.6	15.7	0.776	0.14	-40.7	-2.35	2.88	66%	0.0	33.9
	MinVar-C	10.7	14.5	0.773	0.66	-35.7	-2.15	2.06	0%	3.6	19.5
	MVO-C	12.7	17.5	0.774	0.82	-43.2	-2.64	3.43	0%	6.9	17.5
	HRP-Clip	11.6	15.8	0.775	0.12	-41.1	-2.37	2.80	8%	0.3	34.6
	HRP-Proj	11.7	15.8	0.775	0.11	-41.2	-2.37	2.76	0%	0.3	34.7
	CA-HRP-B	11.9	15.8	0.790	0.55	-41.9	-2.36	2.65	0%	1.0	34.8
	CA-HRP	11.9	15.7	**0.793**	–	-40.4	-2.35	2.57	0%	0.9	34.7

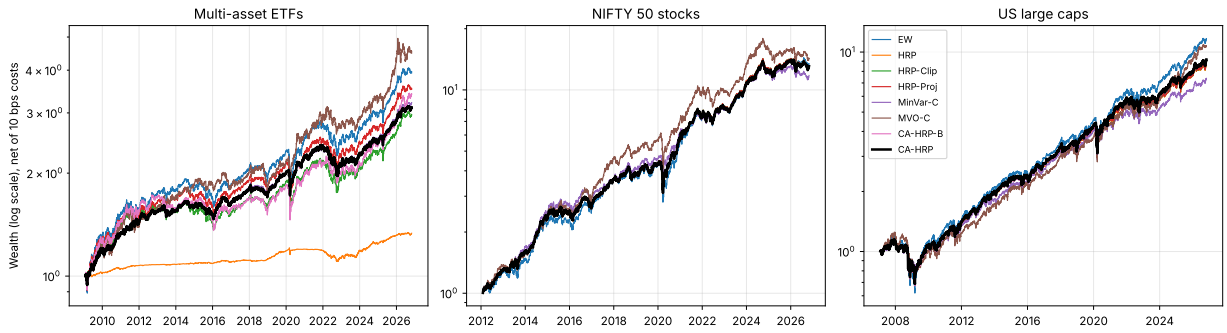


Figure 1: Growth of one unit of wealth (log scale), net of 10 bps costs.

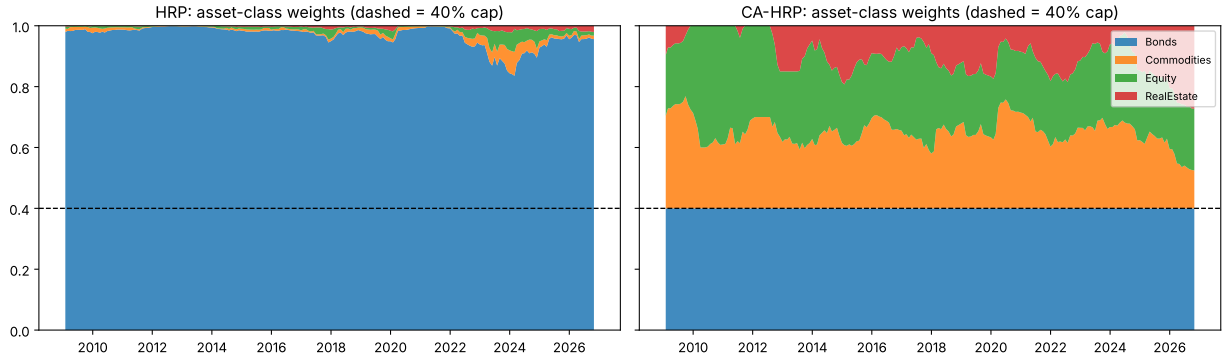


Figure 2: Asset-class weights in the multi-asset universe. Unconstrained HRP (left) is almost entirely in bonds, far above the 40% cap. CA-HRP (right) holds bonds at the cap and allocates the remainder by tracking-error projection.

CA-HRP keeps the bond budget but shifts it from short-term to intermediate and long Treasuries (IEF 13%, TLT 9%), whose negative correlation with equities hedges the risk it must now take, and holds gold (13%), broad commodities (12%) and the S&P 500 (11%) as its main risky positions.

The solver-free CA-HRP-B performs like the Euclidean projection (0.750) but with four times the turnover. Its bisection handles constraints by pushing excess weight down the tree, so the portfolio concentrates in whichever volatile branch has spare capacity (commodities, Figure 2 and Section 4). This is the central negative finding that motivated the final design: constraints are best enforced *after* the risk allocation, in a risk metric, not inside the bisection.

6.3 Performance where constraints rarely bind

In the two stock universes HRP breaches the limits at about half of the rebalances, but only by small amounts: the realised tracking error between HRP and any of its constrained versions is at most 1%. All HRP-based methods are therefore statistically indistinguishable, which is what Proposition 2(b) predicts as the constraints become slack. CA-HRP posts the highest Sharpe ratio of all eleven methods for US large caps (0.793) and ties HRP for NIFTY (1.369 vs 1.368), where constrained minimum variance is marginally higher (1.401, $p = 0.62$). In other words, CA-HRP costs nothing when HRP is already compliant and helps when it is not.

6.4 Held-out test period

Table 3 repeats the comparison on the period after the validation window, which played no role in choosing the design. The ranking is unchanged. In the multi-asset universe CA-HRP's test Sharpe is 0.929, against 0.805 for HRP-Proj, 0.802 for HRP-Clip, 0.783 for equal weight and 0.977 for

constrained minimum variance, with the lowest volatility of any HRP-family method. In stocks CA-HRP again matches HRP.

Table 3: Sharpe ratio (net of 10 bps) on the held-out test period, which was never used to choose the CA-HRP design, next to the full sample. Vol and MaxDD (%) on the test period.

	Test-period Sharpe			ETFs, test period		
Method	ETFs	NIFTY	US	Vol	MaxDD	Full-sample SR
EW	0.783	1.066	0.972	10.6	-24.8	0.763
HRP	0.830	1.126	0.909	2.4	-8.8	0.833
MinVar-C	0.977	1.051	0.889	7.4	-17.5	0.935
MVO-C	0.781	0.869	0.976	12.9	-19.1	0.753
HRP-Clip	0.802	1.126	0.909	8.9	-22.8	0.709
HRP-Proj	0.805	1.126	0.910	9.6	-22.8	0.764
CA-HRP-B	0.876	1.122	0.947	9.5	-21.3	0.750
CA-HRP	0.929	1.123	0.941	7.4	-18.8	0.908

6.5 Ablation

Table 4 isolates each design choice. In the multi-asset universe the *distance metric* drives almost the whole gain: replacing the full-covariance tracking-error metric by its diagonal lowers the Sharpe ratio from 0.908 to 0.777, and the Euclidean metric to 0.742. Adding capacity-aware bisection before the projection lowers it to 0.861 and nearly doubles turnover; adding a tree-consistency penalty has a similar effect. The choice of covariance estimator, linkage rule and split rule matters little (Sharpe 0.910–0.922), so the method is not sensitive to those HRP details. Removing the turnover cap barely changes performance, showing that the cap is rarely binding for CA-HRP and does not drive its results. In the stock universes no ablation moves the Sharpe ratio by more than 0.04.

Table 4: Ablation of CA-HRP components (full sample, net of 10 bps). Each row changes one component of the full method.

	Sharpe ratio			ETFs		
Variant	ETFs	NIFTY	US	Vol	Turn.	TE_{HRP}
CA-HRP (full)	0.908	1.369	0.793	7.4	1.25	6.4
sample covariance	0.922	1.372	0.781	7.3	1.22	6.4
diagonal metric	0.777	1.366	0.791	9.2	1.14	8.5
Euclidean metric	0.742	1.365	0.789	9.8	1.06	9.1
+ capacity bisection	0.861	1.367	0.796	8.5	2.30	7.7
+ tree-consistency term	0.846	1.370	0.794	7.9	2.27	7.1
no turnover cap	0.910	1.369	0.793	7.3	1.26	6.4
Ward linkage	0.910	1.330	0.759	7.4	1.30	6.4
dendrogram splits	0.918	1.392	0.778	7.4	1.19	6.5

6.6 Robustness

Costs and estimation window. Table 5 shows the multi-asset advantage of CA-HRP over the other HRP-family methods survives 50 bps costs (0.840 vs 0.739 for HRP-Proj) and both shorter (126-day) and longer (504-day) estimation windows, where CA-HRP improves on HRP-Proj by 0.14 and 0.19 Sharpe points respectively. Constrained minimum variance remains a strong competitor throughout.

Stability of weights. We re-estimate every method on 30 block-bootstrapped versions of the estimation window at eight dates per universe and measure how much the weights move. CA-HRP is far more stable than mean–variance (instability 0.16–0.22 vs 0.70–0.78) and more stable than constrained minimum variance in every universe (e.g. 0.218 vs 0.324 for NIFTY). In the multi-asset universe HRP-Proj is the most stable, because it is anchored near equal weight; its stability comes at the price of a portfolio that tracks HRP poorly and has higher volatility.

Table 5: Robustness. Left: Sharpe ratio in the multi-asset universe at 50 bps costs and with 126- and 504-day estimation windows (all universes). Right: weight instability, the mean L_1 distance of bootstrapped weights from their average (lower = more stable).

	ETFs	Window 126 days			Window 504 days			Weight instability		
Method	50 bps	ETFs	NIFTY	US	ETFs	NIFTY	US	ETFs	NIFTY	US
EW	0.750	0.535	1.197	0.794	0.701	1.196	0.775	–	–	–
HRP	0.705	0.622	1.288	0.786	0.932	1.287	0.767	0.068	0.230	0.227
MinVar-C	0.868	0.703	1.392	0.801	0.936	1.341	0.730	0.185	0.324	0.318
MVO-C	0.650	0.660	1.142	0.755	0.796	1.172	0.760	0.754	0.698	0.781
HRP-Clip	0.614	0.479	1.293	0.786	0.800	1.288	0.767	0.209	0.222	0.214
HRP-Proj	0.739	0.535	1.293	0.792	0.714	1.288	0.766	0.037	0.220	0.217
CA-HRP	0.840	0.676	1.301	0.803	0.908	1.276	0.766	0.158	0.218	0.211

Stress periods. Table 6 reports cumulative returns in three crises. In the COVID crash CA-HRP lost 6.9% in the multi-asset universe, against 14.0% for HRP-Proj, 13.7% for HRP-Clip and 14.9% for equal weight, and it was flat in the GFC window. The 2022 rates shock is the exception: CA-HRP's hedge tilt towards intermediate and long Treasuries, which protected it in 2020, lost 17.2% when bonds and equities fell together, slightly more than HRP-Proj (−15.3%). This illustrates the main limitation of any covariance-based hedge: it relies on correlations that can break down.

Table 6: Cumulative return (%) in stress windows: GFC = Sep 2008–Mar 2009, COVID = 15 Feb–30 Apr 2020, Rates = Jan–Oct 2022.

	ETFs			NIFTY	US		
Method	GFC	COVID	Rates	COVID	GFC	COVID	Rates
EW	-1.6	-14.9	-16.9	-16.8	-30.4	-13.5	-5.1
HRP	0.6	-0.3	-6.6	-9.6	-24.8	-10.0	-5.9
MinVar-C	0.5	-7.6	-16.0	-4.4	-20.3	-10.7	-4.4
MVO-C	0.8	-4.5	-8.2	-13.8	-30.1	-9.3	-0.9
HRP-Clip	0.4	-13.7	-17.4	-9.6	-24.8	-10.0	-6.3
HRP-Proj	-1.0	-14.0	-15.3	-9.6	-25.0	-10.0	-6.3
CA-HRP-B	-2.2	-11.2	-11.3	-8.4	-25.8	-9.0	-5.3
CA-HRP	0.3	-6.9	-17.2	-8.4	-24.7	-9.5	-5.4

Tightening the constraints. Finally, we tighten the per-asset cap from $4/N$ to $1.25/N$ (with group caps scaled accordingly) and measure, at twelve dates per universe, how far each method's portfolio is from HRP in ex-ante tracking-error terms (Figure 3). Settings that are mathematically infeasible (caps of $1.5/N$ and $1.25/N$ in the multi-asset universe) are detected by a linear feasibility check and skipped; HRP-Clip silently returns infeasible weights in such cases, and even at caps of $2.5/N$ and looser it still breaches limits by 1.1–1.9 percentage points on average in the multi-asset

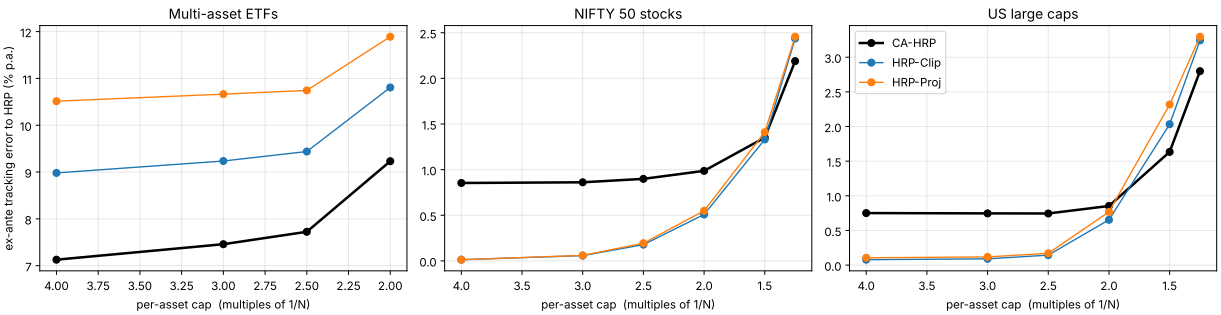


Figure 3: Ex-ante tracking error to sample-covariance HRP as the per-asset cap tightens (right = tighter). Infeasible settings are omitted.

universe. In that universe CA-HRP has the smallest distortion at every cap (7.1–9.2% versus 10.5–11.9% for HRP-Proj) and the lowest volatility. In the stock universes CA-HRP has a small constant floor of about 0.8% because its HRP target uses shrinkage rather than the sample covariance; at the tightest cap ($1.25/N$) it distorts HRP the least in both stock universes (2.2% vs 2.4–2.5% for NIFTY, 2.8% vs 3.2–3.3% for US), and already at $1.5/N$ for US stocks.

7 Discussion

Three lessons emerge. First, *HRP is not deployable as-is* under realistic mandates: it breaches constraints at up to every rebalance, and the clipping fix used in practice still fails often. Second, *how* constraints are enforced matters as much as *whether* they are: the same HRP target, projected with a risk-aware metric instead of a Euclidean one, gives a portfolio with 25% less volatility and a higher Sharpe ratio, out of sample and in the held-out period. Third, building constraints into the bisection, while elegant and provably feasible, is inferior to separating risk allocation from constraint enforcement.

Limitations. The stock universes are survivorship-biased because they contain current large caps; this affects all methods equally. The gains of CA-HRP are concentrated in settings where constraints bind strongly; when they do not, it reduces to HRP, which is by design. Most Sharpe differences are not statistically significant at conventional levels, which is typical for portfolio comparisons over 15–20 years, and constrained minimum variance remains a strong benchmark. Finally, the tracking-error projection inherits the weakness of any covariance-based hedge, visible in 2022 when stock–bond correlation turned positive.

8 Conclusion

We showed that Hierarchical Risk Parity, despite its robustness, cannot be used directly under realistic portfolio mandates, and that the ad hoc fixes used in practice either fail to restore feasibility or distort the risk allocation. Constraint-Aware HRP separates the two problems: HRP decides how risk should be allocated, and a tracking-error projection finds the compliant portfolio closest to that allocation in risk terms. The method has clean theoretical properties, runs in milliseconds per rebalance, satisfied every constraint in more than 600 rebalances across three markets, and improved risk-adjusted returns exactly where constraints bind, including on a held-out test period. We also proved that constraints can be embedded directly in the recursive bisection with an exact feasibility guarantee, but found that doing so is empirically inferior, a useful negative result for the growing literature on hierarchical allocation.

Future work includes projection metrics that adapt to regime changes in correlations, such as the 2022 stock–bond shift, the use of CA-HRP with factor-model covariance estimates for larger

universes, and extending the capacity analysis to non-linear constraints such as tracking-error or CVaR budgets.

Code and data availability. All code, data and scripts that reproduce every table and figure are available at `https://github.com/pramukhjain123/capstone`.

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A Design study on the validation period

Table 7 lists the eight configurations compared before the method was frozen, using only data up to the end of the validation window. The tracking-error projection was best in the multi-asset universe and tied for best in the stock universes. The sample-covariance version scored marginally higher than the Ledoit–Wolf one; we kept Ledoit–Wolf, chosen beforehand for its better conditioning, since the difference is within noise. The initial design ("cap+tree (v0)") was the weakest in the multi-asset universe.

Table 7: Design study on the validation period only (net of 10 bps). "cap" = capacity-aware bisection; "tree" = tree-weighted projection; "TE" = tracking-error projection. LW covariance unless stated.

	Validation Sharpe			ETFs	
Variant	ETFs	NIFTY	US	Vol	Turn.
cap+tree (v0)	0.605	1.737	0.639	10.2	2.59
euclid (LW)	0.675	1.731	0.645	10.2	1.07
cap+euclid	0.637	1.737	0.646	10.3	2.11
diag-var proj	0.743	1.731	0.648	9.5	1.13
TE proj	0.878	1.738	0.651	7.3	1.19
cap+TE proj	0.750	1.737	0.651	8.6	2.45
TE+tree proj	0.749	1.741	0.651	7.9	2.30
TE proj (sample cov)	0.887	1.745	0.655	7.3	1.11

B Turn-of-month returns in the NIFTY universe

Daily returns on the last trading day of the month are unusually large in the NIFTY sample (an average of 0.42% for an equal-weight portfolio up to 2018, against 0.07% on other days). All methods trade at the month-end close and earn that day's return with their pre-trade weights, so the effect raises all NIFTY Sharpe ratios similarly and does not change their ranking.